



INFERENCE STATISTICS – BUSINESS REPORT

Barnabas R - PGDDS

Table of Contents

1. Introduction and Business Context

2. Problem 1: Foot Injuries and Player Positions

- * 2.1 Problem Statement and Context
- * 2.2 Approach
- * 2.3 Analysis, Inferences, and Insights
 - * 1.1 Probability of a Randomly Chosen Player Suffering an Injury
 - * 1.2 Probability that a Player is a Forward or a Winger
 - * 1.3 Probability that a Randomly Chosen Player Plays in a Striker Position and Has a Foot Injury
 - * 1.4 Probability that a Randomly Chosen Injured Player is a Striker

3. Problem 2: Gunny Bag Breaking Strength

- * 3.1 Problem Statement and Context
- * 3.2 Approach
- * 3.3 Analysis, Inferences, and Insights
 - * 2.1 Proportion of Gunny Bags with Breaking Strength Less than 3.17 kg per sq cm
 - * 2.2 Proportion of Gunny Bags with Breaking Strength at Least 3.6 kg per sq cm
 - * 2.3 Proportion of Gunny Bags with Breaking Strength Between 5 and 5.5 kg per sq cm
 - * 2.4 Proportion of Gunny Bags with Breaking Strength NOT Between 3 and 7.5 kg per sq cm

4. Conclusion

5. Problem 3: Zingaro Stone Printing Hardness Analysis

- * 5.1 Problem Statement and Context
- * 5.2 Approach
- * 5.3 Analysis, Inferences, and Insights
 - * 3.1 Zingaro's Belief Regarding Unpolished Stones

- * 3.2 Comparison of Mean Hardness Between Polished and Unpolished Stones

6. Problem 4: Dental Implant Hardness Analysis

- * 6.1 Problem Statement and Context
- * 6.2 Approach
- * 6.3 Analysis, Inferences, and Insights
 - * 4.1 Variation in Hardness by Dentists
 - * 4.2 Variation in Hardness by Methods
 - * 4.3 Interaction Effect Between Dentist and Method for Each Alloy
 - * 4.4 Hardness Variation Depending on Dentists and Methods Together (Full Model)

7. Overall Conclusions and Recommendations

Figure Numbers

- **Figure 2.1: Histogram with the area below 3.17 kg/sq cm shaded**
- **Figure 2.2: Histogram with the area at or above 3.6 kg/sq cm shaded**
- **Figure 2.3: Histogram with the region between 5 and 5.5 kg/sq cm shaded**
- **Figure 2.4: Histogram with the regions below 3 kg/sq cm and above 7.5 kg/sq cm shaded**
- **Figure 3.1: Histogram showing the distribution of unpolished stone hardness with a vertical line at 150**
- **Figure 3.2: Boxplot comparing the Brinell Hardness Index of polished and unpolished stones**
- **Figure 4.1: Boxplot illustrating hardness by Dentist**
- **Figure 4.2: Boxplot illustrating hardness by Method**
- **Figure 4.3: Interaction plot for Alloy 1**
- **Figure 4.4: Interaction plot showing Method by Dentist across all alloys**

1. Introduction and Business Context

This report presents a statistical analysis addressing four distinct business problems. The full scope of this analysis covers a broader range of operational insights. These problems involve understanding the relationship between foot injuries and player positions in a male football team, assessing the breaking strength characteristics of gunny bags for cement packaging, analyzing the hardness of Zingaro stones for printing, and investigating factors influencing dental implant hardness. The insights derived from this analysis aim to support informed decision-making for a physiotherapist in a sports context, a quality team in a manufacturing/supply chain context, a printing company, and a dental quality team, respectively. The report prioritizes the business context, problem, solution mechanism, analysis, and recommendations for all stakeholders, including those without a coding background.

2. Problem 1: Foot Injuries and Player Positions

2.1 Problem Statement and Context:

A physiotherapist for a male football team is interested in understanding the relationship between foot injuries and the positions at which players play. This analysis aims to help identify potential risk factors or prevalence rates associated with different positions, informing injury prevention strategies.

2.2 Approach To address this problem:

The analysis involved calculating basic probabilities based on collected data regarding player positions (Striker, Forward, Attacking Midfielder, Winger) and their injury status (Injured, Not Injured). These probabilities provide insights into the likelihood of injuries or specific player types within the team.

2.3 Analysis, Inferences, and Insights Based on the probability calculations, the following insights were derived:

- Probability of a Randomly Chosen Player Suffering an Injury:

The probability that a randomly chosen player would suffer an injury is 0.617, or 61.7%. This indicates a relatively high overall injury rate among the players in the dataset collected.

- Probability that a Player is a Forward or a Winger:

The probability that a player is either a forward or a winger is 0.523, or 52.3%. This suggests that these two positions combined account for a little over half of the players in the dataset.

- Probability that a Randomly Chosen Player Plays in a Striker Position and Has a Foot Injury:

The probability that a randomly chosen player plays in a striker position and has a foot injury is 0.191, or 19.1%. This is the joint probability of a player being both a striker and injured.

- Probability that a Randomly Chosen Injured Player is a Striker:

The probability that a randomly chosen injured player is a striker is 0.310, or 31.0%. This is a conditional probability, indicating that among the injured players, nearly one-third are strikers. This insight is particularly valuable for the physiotherapist, highlighting strikers as a significant proportion of the injured players, potentially warranting closer attention or specific preventative measures for this position.

3. Problem 2: Gunny Bag Breaking Strength

3.1 Problem Statement and Context:

The quality team of a cement company is concerned with understanding the breaking strength of gunny bags used for packaging cement to better manage wastage or pilferage within the supply chain. The breaking strength is known to be normally distributed with a mean of 5 kg per sq. centimeter and a standard deviation of 1.5 kg per sq. centimeter. The team seeks to determine the proportion of bags that fall within or outside specific strength ranges.

3.2 Approach:

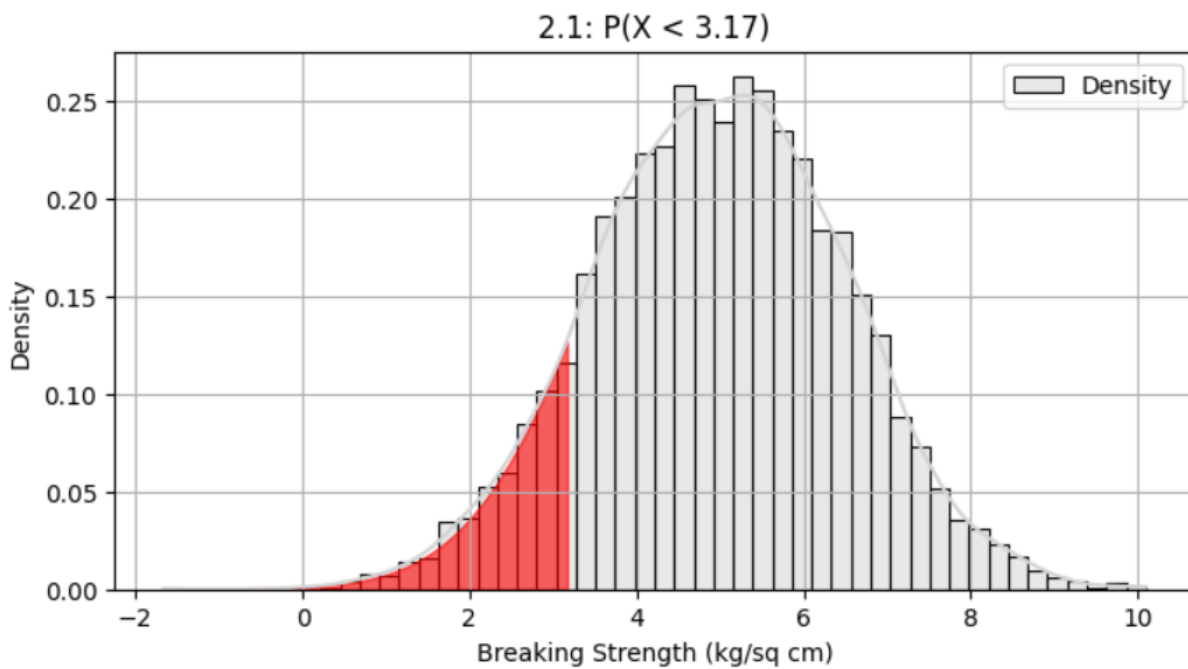
To determine the proportions of gunny bags within specified breaking strength ranges, the analysis utilized the properties of the normal distribution. Specifically, the cumulative distribution function (CDF) of the normal distribution was employed to calculate probabilities for values less than a certain point, greater than a certain point, or between two points. The analysis also involved simulating a large number of samples (10,000) from this normal distribution to support visualizations.

3.3 Analysis, Inferences, and Insights The analysis provided the following proportions and associated insights:

- Proportion of Gunny Bags with Breaking Strength Less than 3.17 kg per sq cm:

Approximately 11.12% of the gunny bags exhibit a breaking strength below 3.17 kg per sq cm. This proportion represents bags that are significantly weaker than the mean, indicating a

percentage of bags that might be prone to breaking prematurely, leading to wastage. The corresponding Z-score is -1.22.

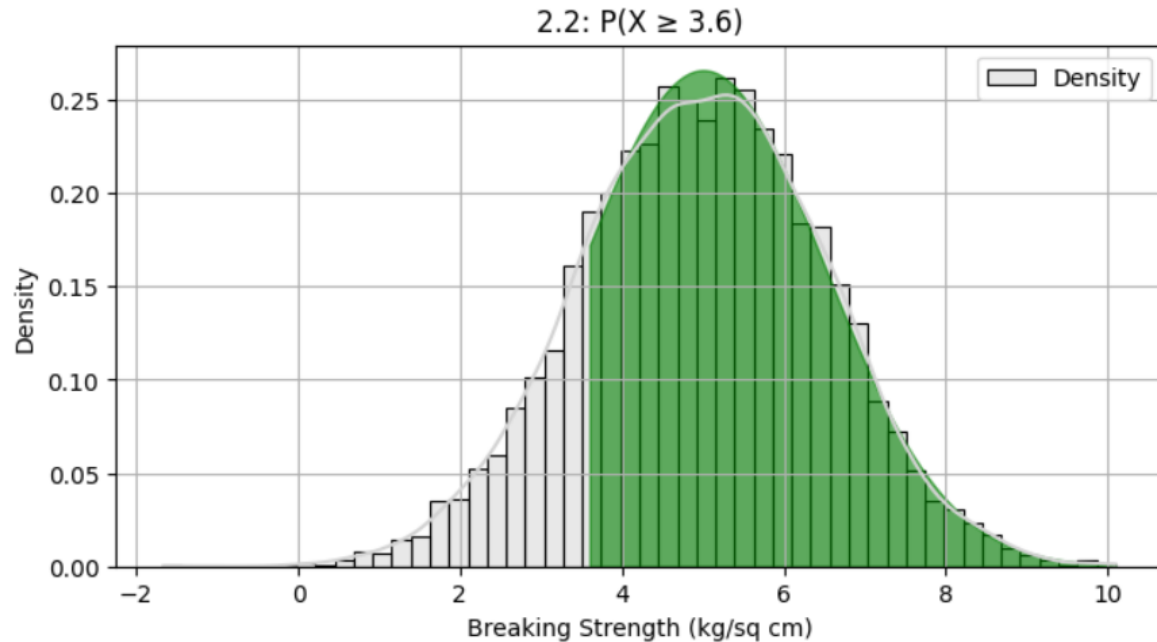


2.1: $P(X < 3.17) = 0.1112$

Fig: 2.1

- Proportion of Gunny Bags with Breaking Strength at Least 3.6 kg per sq cm:

About 82.47% of the gunny bags possess a breaking strength equal to or exceeding 3.6 kg per sq cm. This indicates that a large majority of the bags meet or exceed a strength threshold, suggesting good overall quality for most packaging material. The Z-score is -0.93.

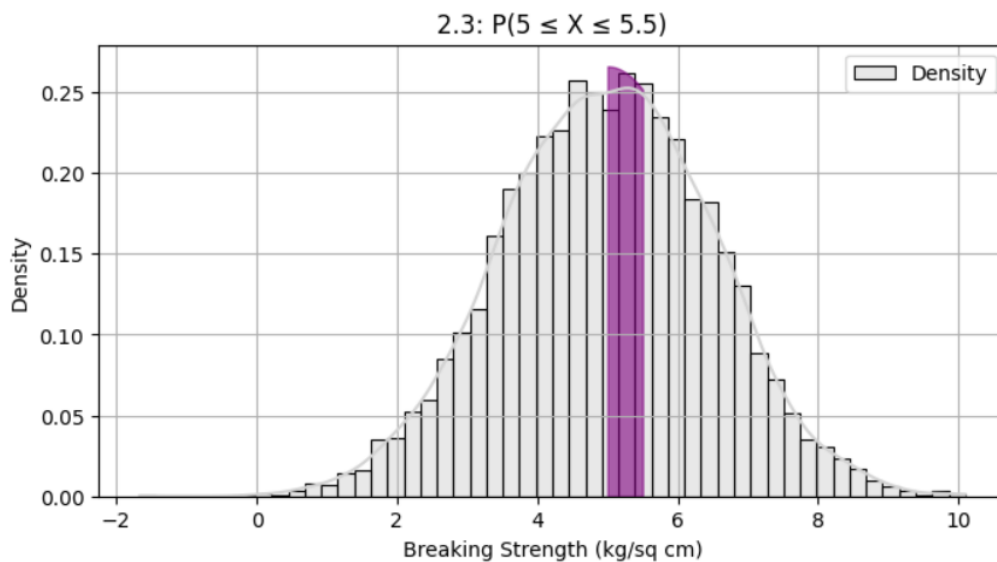


2.2: $P(X \geq 3.6) = 0.8247$

Fig: 2.2

- Proportion of Gunny Bags with Breaking Strength Between 5 and 5.5 kg per sq cm:

The proportion of gunny bags with a breaking strength falling between 5 and 5.5 kg per sq cm is around 13.06%. This range is centered around the mean (5 kg/sq cm). The Z-scores are 0.00 and 0.33, respectively.

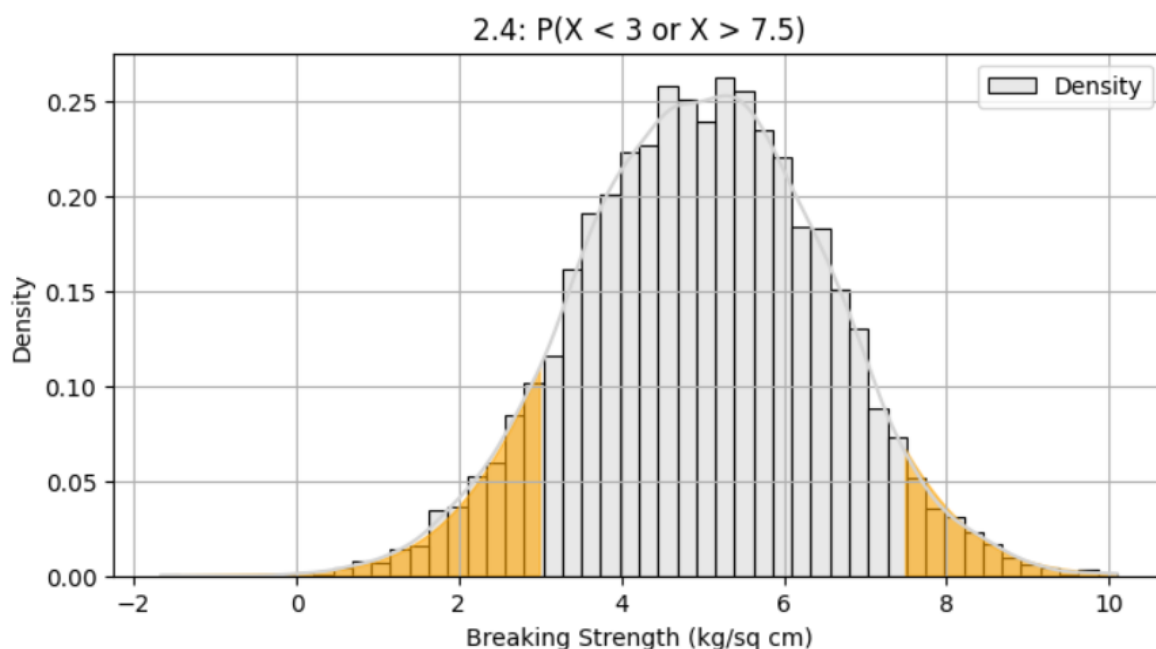


2.3: $P(5 \leq X \leq 5.5) = 0.1306$

Fig: 2.3

- Proportion of Gunny Bags with Breaking Strength NOT Between 3 and 7.5 kg per sq cm:

Roughly 13.90% of the gunny bags have a breaking strength that falls outside the range of 3 to 7.5 kg per sq cm. This proportion represents bags that are either significantly too weak (below 3 kg/sq cm) or exceptionally strong (above 7.5 kg/sq cm). The Z-scores are -1.33 and 1.67, respectively. This insight is crucial for the quality team, as these are the bags most likely to be problematic (too weak, leading to pilferage/wastage) or potentially over-engineered (too strong, leading to unnecessary material cost if not required)



2.4: $P(X < 3 \text{ or } X > 7.5) = 0.139$

Fig: 2.4

4. Problem 3: Zingaro Stone Printing Hardness Analysis

4.1 Problem Statement and Context:

Zingaro stone printing requires a Brinell's hardness index of at least 150 for optimal image printing on stone surfaces. The company has received a new batch of polished and unpolished stones and has concerns about the suitability of unpolished stones. The quality team needs to confirm whether these unpolished stones meet the minimum hardness requirement and if

there's a difference in mean hardness between polished and unpolished stones. This analysis assumes a 5% significance level.

4.2 Approach To address these concerns, statistical hypothesis testing was employed:

- For the suitability of unpolished stones, a one-sample t-test was conducted to determine if the mean hardness of unpolished stones is significantly less than the required minimum of 150.
- For comparing the mean hardness of polished and unpolished stones, a paired t-test was performed, assuming the samples were related. An independent two-sample t-test was also performed as an alternative comparison. These tests assess if there is a statistically significant difference between the means of the two stone types.

4.3 Analysis, Inferences, and Insights:

- Zingaro's Belief Regarding Unpolished Stones: Zingaro is justified in believing that the unpolished stones may not be suitable for printing.
 - A one-sample t-test yielded a t-statistic of -4.1646 and a one-tailed p-value of 0.0.
 - Since the p-value (0.0) is less than the assumed significance level of 0.05, the null hypothesis (mean hardness is 150 or greater) is rejected.
 - This statistically significant result indicates that the mean Brinell hardness of the unpolished stones is significantly below 150. This finding strongly supports Zingaro's concern that unpolished stones in this batch are likely to be below the optimal hardness for printing, which could lead to suboptimal print quality or increased waste.

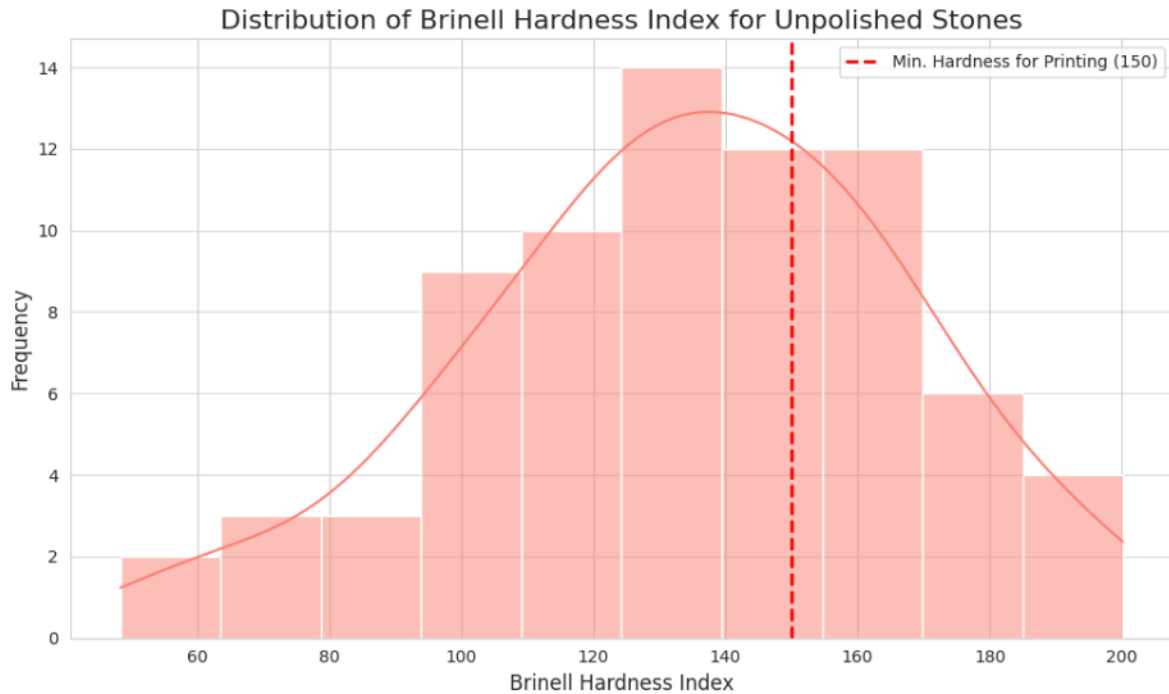


Fig: 3.1

- Comparison of Mean Hardness Between Polished and Unpolished Stones: The mean hardness of the polished and unpolished stones is not the same.
 - A paired t-test resulted in a t-statistic of -3.5589 and a two-tailed p-value of 0.0007. Since this p-value (0.0007) is less than the 0.05 significance level, the null hypothesis (that the mean hardness is the same) is rejected, indicating a statistically significant difference.
 - Furthermore, an independent two-sample t-test yielded a T-statistic of -3.2422 and a P-value of 0.0016, also leading to the conclusion to reject the null hypothesis, confirming that the mean hardness of polished and unpolished stones is not the same.
 - This implies that the polishing process has a significant impact on the stone's hardness, likely increasing it. This insight is critical for Zingaro in selecting appropriate stone types for different printing requirements

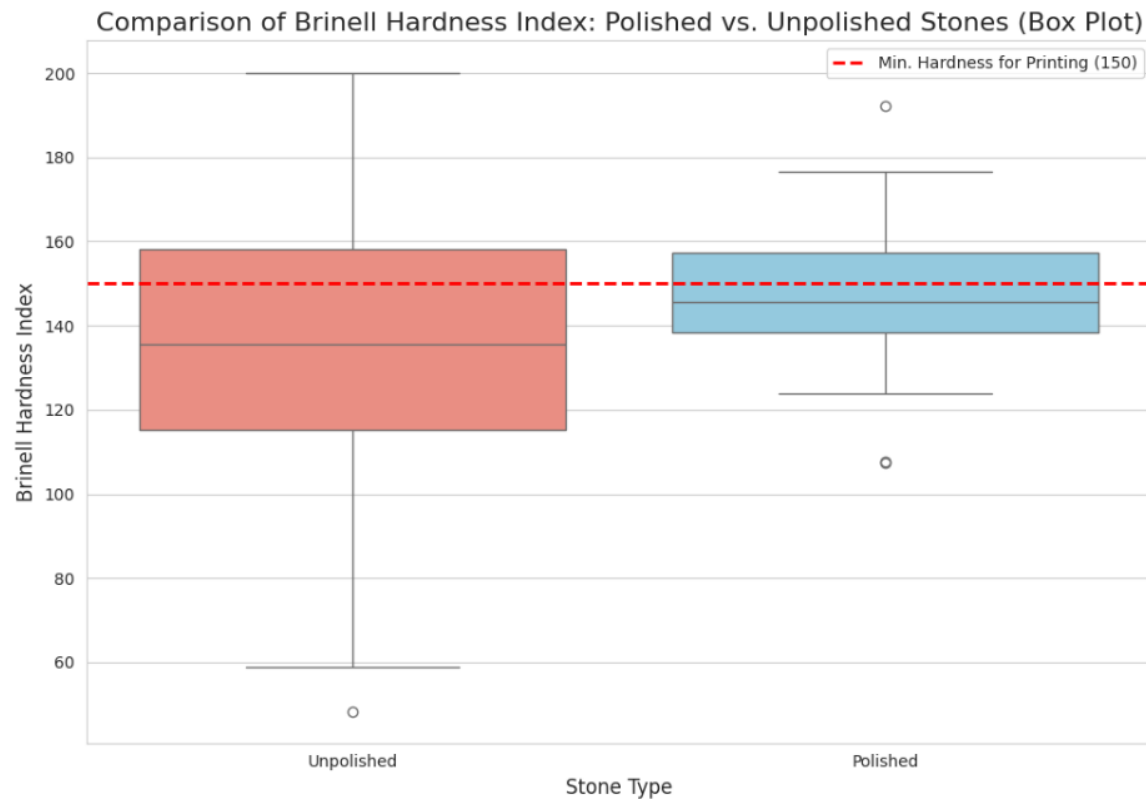


Fig 3.2

5. Problem 4: Dental Implant Hardness Analysis

5.1 Problem Statement and Context:

The hardness of metal implants in dental cavities is influenced by multiple factors, including the implant method, treatment temperature, alloy used, and individual dentists. The quality team wants to understand how these factors, individually and in combination, affect the implant hardness (the 'Response' variable). This analysis will help optimize implant procedures and ensure consistent quality.

5.2 Approach To analyze the impact of multiple categorical factors on implant hardness, Analysis of Variance (ANOVA) techniques were employed:

- One-Way ANOVA was used to assess the individual effect of 'Dentist' and 'Method' on hardness.
- Two-Way ANOVA with interaction was performed for each 'Alloy' type to identify combined effects of 'Dentist' and 'Method' specific to that alloy.

- A Full Two-Way ANOVA incorporating all factors ('Dentist', 'Method', 'Alloy', and 'Temp') and their interactions was conducted to provide a comprehensive understanding of their collective influence on hardness.

- Visualizations such as boxplots and interaction plots were used to support the ANOVA findings.

5.3 Analysis, Inferences, and Insights

- Variation in Hardness by Dentists: The hardness of implants does not significantly vary depending on dentists when considered as a single factor.

- A One-Way ANOVA for 'Response' by 'Dentist' yielded an F-value of 1.9345 and a p-value ($PR(>F)$) of 0.1121. Since the p-value is greater than 0.05, we fail to reject the null hypothesis, suggesting no statistically significant difference in implant hardness across different dentists in this dataset based on this specific test.

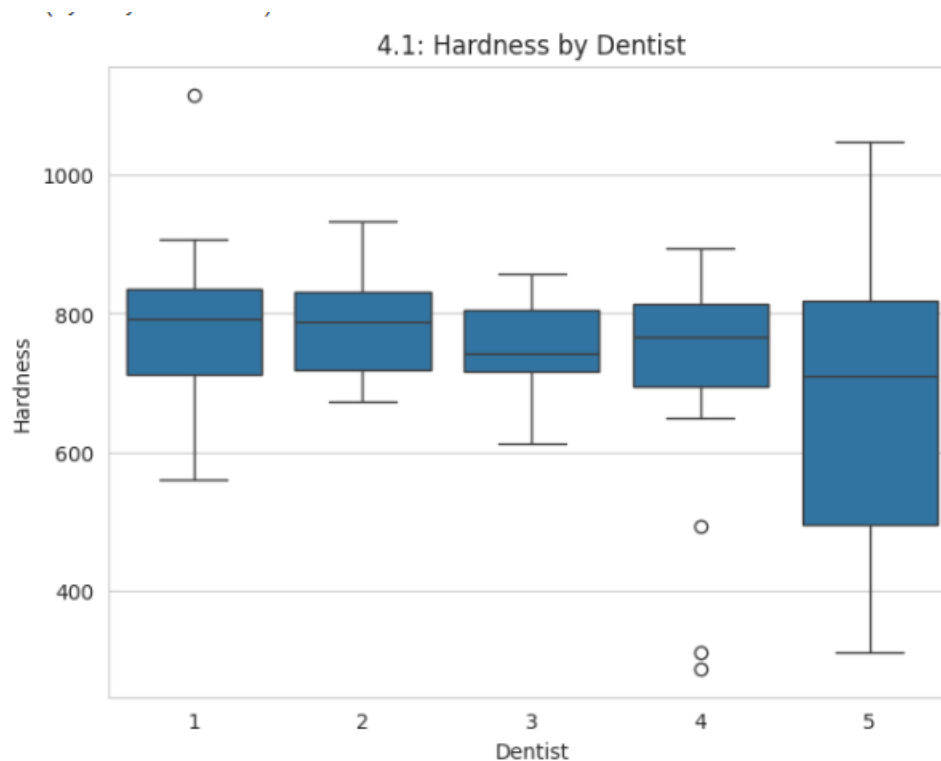


Fig 4.1

- Variation in Hardness by Methods: The hardness of implants significantly varies depending on the methods used.

- A One-Way ANOVA for 'Response' by 'Method' yielded an F-value of 19.8927 and a p-value ($PR(>F)$) of 0.0. Since the p-value is less than 0.05, we reject the null hypothesis, indicating a

statistically significant difference in implant hardness across different methods. This is a crucial finding, suggesting that the choice of implant method directly impacts the resulting hardness.

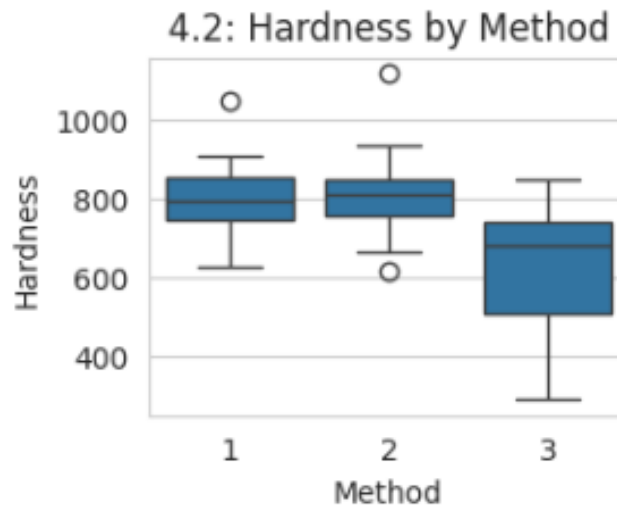


Fig 5.2

- Interaction Effect Between Dentist and Method for Each Alloy: There is a significant interaction effect between Dentist and Method on hardness for Alloy 1, but not for Alloy 2.

- For Alloy 1: The interaction term has an F-value of 3.3984 and a p-value of 0.0068. Since this p-value is less than 0.05, there is a significant interaction effect. This means that for Alloy 1, the effect of the 'Method' on hardness is dependent on the 'Dentist', and vice versa.

- For Alloy 2: The interaction term has an F-value of 1.9228 and a p-value of 0.0932. Since this p-value is greater than 0.05, there is no statistically significant interaction effect. This means that for Alloy 2, the effect of 'Method' on hardness is generally consistent across different 'Dentists'. This distinction between alloys is vital for the quality team.

4.3: Interaction Plot for Alloy 1

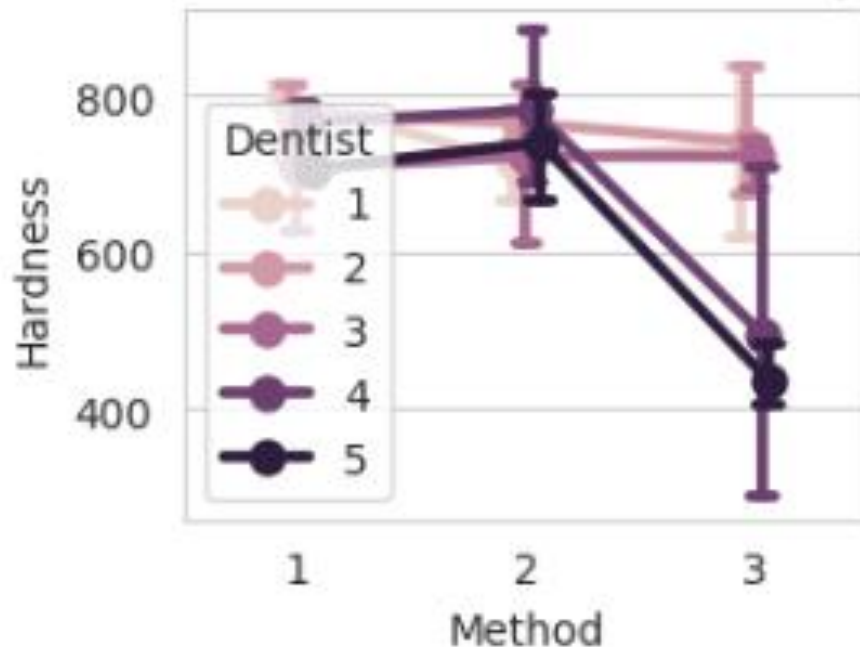


Fig 5.3

- Hardness Variation Depending on Dentists and Methods Together (Full Model): When considering all factors together, the hardness of implants significantly varies based on Dentists, Methods, Alloy, and several interaction terms.

- A Full Two-Way ANOVA including C(Dentist), C(Method), C(Alloy), and their interactions yielded the following significant effects (p-value < 0.05):

- C(Dentist): p-value = 0.0060 (Significant main effect). While individually not significant, Dentists do show a significant influence in the full model.

- C(Method): p-value = 0.0000 (Highly significant main effect).

- C(Alloy): p-value = 0.0017 (Significant main effect).

- C(Dentist):C(Method): p-value = 0.0009 (Significant interaction). This confirms a significant combined effect of Dentist and Method across all alloys.

- Other interactions, such as C(Dentist):C(Alloy) and C(Dentist):C(Method):C(Alloy), were not statistically significant at 0.05.

- These findings underscore the complexity of dental implant hardness. The Dentist, Method, and Alloy type are all crucial main factors, and importantly, there is a significant interaction

between the Dentist and the Method, meaning the choice of method may affect hardness differently depending on the dentist, even across all alloys.

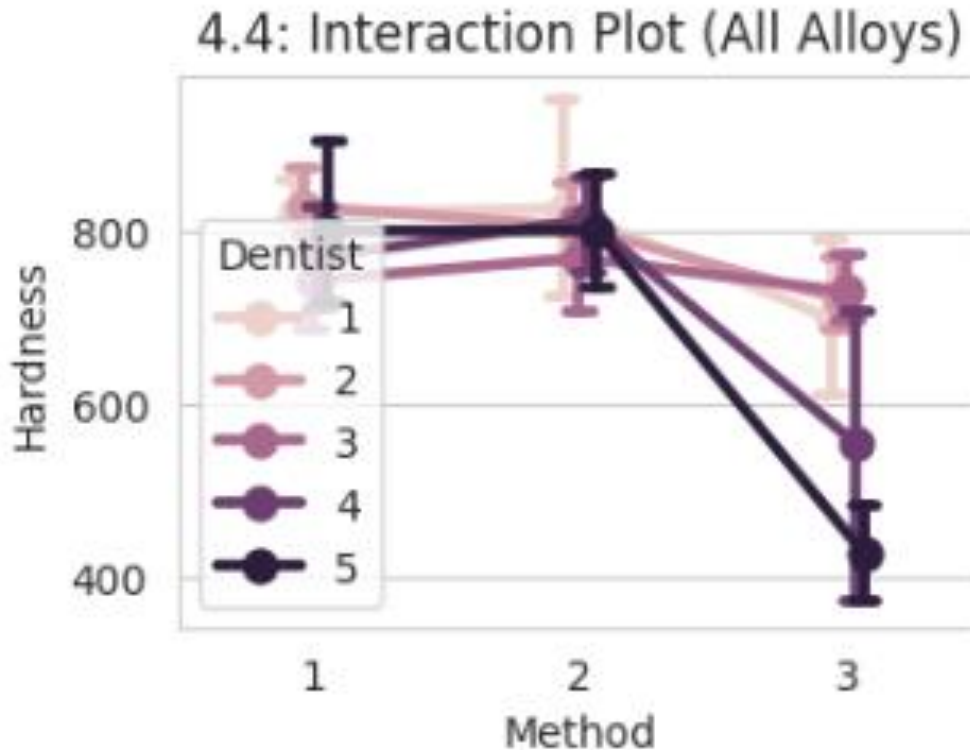


Fig 5.4

6. Overall Conclusions and Recommendations

This report has provided critical statistical insights for the football team's injury management, the cement company's packaging quality control, Zingaro stone printing, and dental implant hardness analysis.

• For the Physiotherapist (Problem 1):

The high overall injury rate (61.7%) and the significant proportion of injured strikers (31.0%) suggest that strikers might be a high-risk group requiring targeted injury prevention and rehabilitation programs.

• For the Cement Quality Team (Problem 2):

A noticeable proportion of bags (11.12%) are weaker than 3.17 kg/sq cm, which could lead to wastage. The quality team should focus on the 13.90% of bags falling outside the 3 to 7.5 kg/sq

cm range to minimize both excessively weak bags (pilferage risk) and potentially over-engineered bags (cost inefficiency). Regular quality checks and supplier engagement are recommended.

• **For Zingaro Stone Printing (Problem 3):**

The unpolished stones are indeed unsuitable for printing based on hardness, and there's a significant difference in hardness between polished and unpolished stones. Zingaro should primarily use polished stones for critical printing tasks or consider pre-treatment processes for unpolished stones to achieve the minimum hardness.

• **For Dental Implant Quality (Problem 4):**

- The Method of implant is a highly significant factor influencing hardness, suggesting that standardizing and optimizing the chosen method can significantly improve consistency.
- There is a significant interaction between the Dentist and the Method, implying that the effectiveness of a particular method can vary depending on the dentist.
- The Alloy used also significantly impacts hardness.
- Recommendations include developing method-specific training for dentists, potentially tailoring methods to individual dentist strengths for Alloy 1, and establishing clear guidelines for optimal method-alloy combinations to ensure consistent implant hardness across the board.

This updated report serves as a foundation for data-driven decisions aimed at improving operational efficiency, product quality, and risk management within these distinct business contexts.

-----XXXX-----